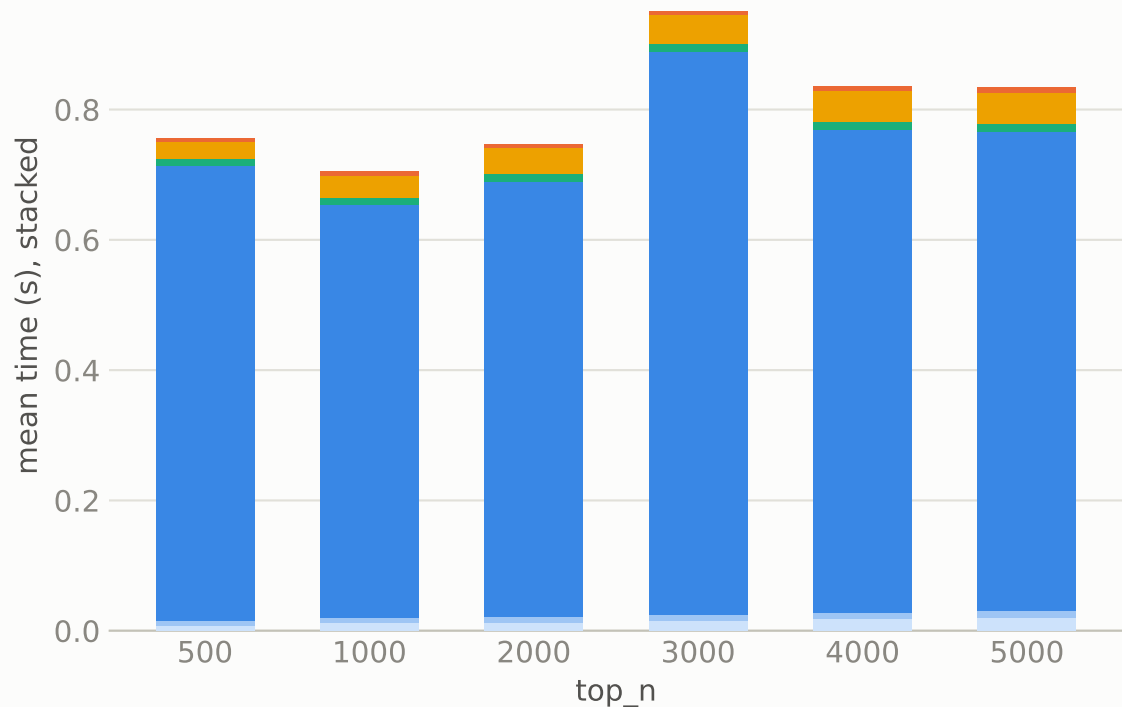


tier_100k: stage-time breakdown vs. top_n ($\sigma=0.30$, mean over α, k)



Probe (HNSW + overlap)

Union combine

N_i/n_i lookup

Stage 1 (Dinkelbach)

Unionability scoring

Stage 2 (ILP)